

DAWATRACK: Your Personalized Medicine Guardian

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Abstract - DawaTrack is a next-generation, IoT-enabled digital health ecosystem that integrates real-time physiological monitoring, predictive analytics, and intelligent alert mechanisms to facilitate proactive healthcare management. Built around the ESP32 microcontroller and MAX30100 sensor, the system continuously records timestamped vitals—namely heart rate and blood oxygen saturation (SpO₂)—and processes them via a hybrid model combining fuzzy logic and XGBoost classification for personalized risk assessment. At its core lies a dynamic web-based Sensor Dashboard, which visualizes live and historical data with intuitive indicators such as “Normal,” “Bradycardia,” “Tachycardia,” “Caution,” and “Danger.” The system issues automated alerts with advisory messages via email when anomalies are detected. The system’s robust architecture, dual-layered feedback mechanism, and cloud-based integration make it a promising solution for remote healthcare, rural areas, and elderly patient management.

Keywords — IoT in Healthcare, Real-Time Monitoring, Heart Rate, SpO₂, Predictive Analytics, Fuzzy Logic, XGBoost, ESP32, Supabase, Wearable Sensors, Preventive Healthcare, Intelligent Alert Systems.

I. INTRODUCTION

As healthcare is becoming more data-driven and patient-centric, real-time monitoring of physiological signals is becoming more and more of a requirement—especially in chronic disease management, geriatric care, and rural care. The conventional model of healthcare depends on episodic physician visits, which seldom cover short-term fluctuations of vital signs that presage serious diseases.

DawaTrack fills this void through integrating wearable sensing hardware, smart backend analytics, and interactive web dashboard. It not only shows heart rate and SpO₂ readings, but it also uses a mix of fuzzy logic rules and XGBoost machine learning models to make sense of them. Then, these interpretations are turned into personalized health alerts that are sent by email so that users can act quickly. DawaTrack is unique in that it includes user feedback loops, real-time alerts, predictive suggestions, and modules that will be compatible with mobile and AI in the near future. hi this

The suggested system is deployed using an inexpensive and solid hardware configuration of the ESP32 microcontroller and MAX30100 sensor module to provide data in real-time.

The gathered data is streamed onto a web-based Sensor

Dashboard, where it is converted into beneficial decisions through beneficial visualizations and multi-level status grades like "Bradycardia," "Caution," and "Danger.". Underlying this interface is a robust analysis engine that leverages a combination of XGBoost and fuzzy logic models to evaluate risk and initiate smart alerts with advisory notifications. The following article outlines the whole design and methodology of the DawaTrack ecosystem, from its hardware base and cloud platform to predictive capabilities and vision for the future, illustrating an integrated solution to advanced preventive health.

II. LITERATURE REVIEW

A. Sharma & R. Gupta (2024) [1] emphasized the critical need for continuous heart monitoring systems, particularly for elderly patient care.

R. Ballinger et al. (2018) [2] presented DeepHeart, a deep learning framework designed for cardiovascular risk detection on mobile platforms.

A. Wardana (2022) [3] demonstrated the effectiveness of using fuzzy logic models for the specific application of detecting hypoxemia.

L. Wang & M. Zhou (2023) [4] highlighted the use of SpO₂ data as a basis for respiratory prediction, showcasing its utility in predictive health.

P. Deshmukh et al. (2025) [5] explored the application of the XGBoost machine learning model for the task of general health risk detection.

S. Singh et al. (2025) [6] linked the use of smart health GUI dashboards to increased health literacy and higher user engagement.

The Supabase Docs (2024) [7] describe the technical foundation of the platform as a real-time Postgres backend suitable for modern web applications.

B. Kumar (2023) [8] provides a broad overview of the role and applications of the Internet of Things (IoT) in the field of healthcare monitoring.

M. Kale et al. (2022) [9] discussed the specific use of wearable technology as a tool for monitoring patients with chronic

conditions.

A. Yadav (2021) [10] provided an example of implementing SpO₂ alert systems using the Arduino microcontroller platform.

K. Patel (2023) [11] reinforced the viability of using fuzzy models as a method for performing risk assessment in health applications.

N. Chatterjee (2024) [12] highlighted the effectiveness of the XGBoost algorithm in the context of analyzing complex biomedical data.

The World Health Organization (2021) [13], in its World Health Report, underscored the general importance of continuous health monitoring.

T. Rajan et al. (2023) [14] detailed the use of the ESP32 microcontroller specifically within the domain of biomedical applications. N. Thomas (2022)

[15] discussed the use of advanced AI models for the purpose of predicting vital signs.features.

The DawaTrack system was created to address the fragmented nature of research in IoT health monitoring. While many cited studies make valuable contributions, they often concentrate on just one part of the problem. For instance, research by Wardana, Deshmukh, and Chatterjee emphasizes analytical models like Fuzzy Logic or XGBoost, but they do not offer a complete user-facing implementation with real-time feedback. Similarly, Singh et al. highlight the need for a GUI dashboard but may miss the importance of the predictive engine. In contrast, Rajan et al. and Yadav focus on hardware components like ESP32 or Arduino without considering the larger picture. This specialization leaves a gap between theoretical models or individual parts and a practical, connected system. DawaTrack bridges this gap by creating a complete end-to-end ecosystem. It brings together strong hardware, smart hybrid analytics, and an easy-to-use multi-level classification dashboard into one unified unit.

III. METHODOLOGY

To address the persistent challenges in medication adherence, fragmented health records, and lack of real-time vitals monitoring, DawaTrack is developed as an integrated solution that merges AI-driven insights with blockchain-backed data integrity. The system is structured to bridge gaps in timely health interventions by offering accurate fruit grading for nutritional planning, seamless wearable syncing for continuous monitoring, and tamper-proof medical data storage for enhanced traceability. This methodology outlines the step-by-step development and integration of DawaTrack's intelligent modules aimed at solving these critical pain points.

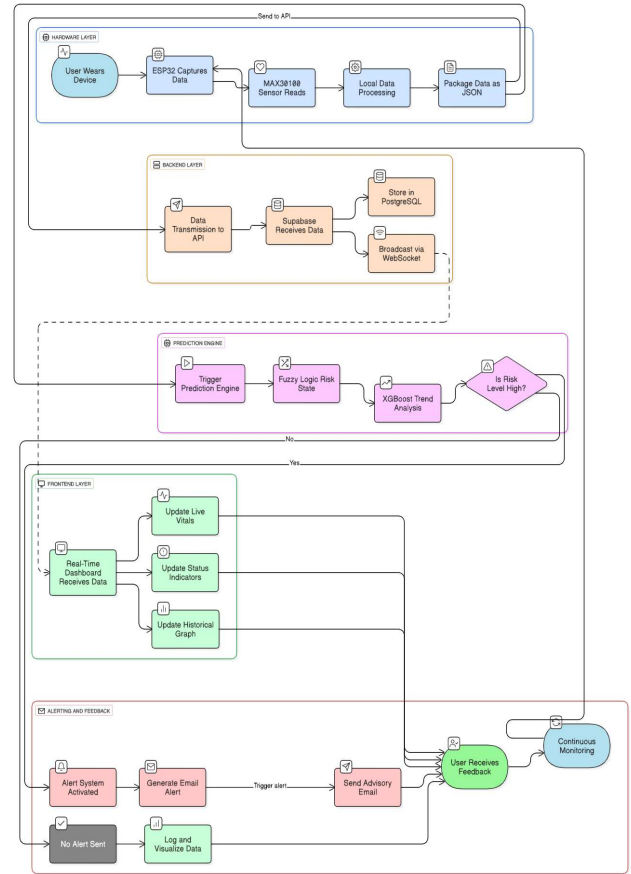


Fig. 1: Workflow of the DawaTrack Health Monitoring System

The DawaTrack system emphasizes proactive healthcare. It uses real-time risk prediction and early detection of issues. The process begins with the hardware layer, which features an ESP32-based wearable device. This device includes an optical sensor that captures raw photoplethysmography (PPG) data from the user. The ESP32's processing unit runs algorithms on this raw data to calculate heart rate (BPM) and blood oxygen saturation (SpO₂). Each calculated data point is stored in a JSON payload. This payload contains a user identifier and an exact timestamp. It is then sent securely through an HTTPS POST request to a RESTful API endpoint managed by the Supabase backend.

MAX30102_Heart Rate and Pulse Oximeter Sensor Using ESP32

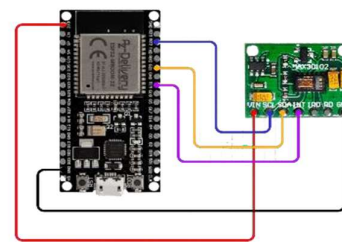


Fig. 2: IoT-Based Vital Sensing Unit built using an ESP32 microcontroller and MAX30100 sensor.

Once the backend receives the data, it processes it in parallel for storage and real-time distribution. The payload is ingested and added as a new entry in a PostgreSQL database, creating a

lasting historical record of the user's vitals. This database transaction is an important event that triggers Supabase's real-time engine, which quickly sends the new data packet over an active WebSocket channel. This keeps any subscribed clients updated with very little delay; this is essential for the live dashboard functionality.

The main user interface, called the "Sensor Dashboard," maintains a steady WebSocket connection to the backend. It listens for real-time data broadcasts. When it receives a new data packet, the client-side JavaScript framework updates the application's state. This change triggers an immediate re-rendering of the UI components. You can see this as the dashboard's gauges refresh to show the latest readings, such as 50 BPM or 200 BPM. The status indicators change color and text to provide instant context. The system's multi-tiered classification shows readings like 94% SpO₂ as "Caution" and 50% as "Danger." The "Readings History" graph also adds the new data point and visually charts physiological trends over time.

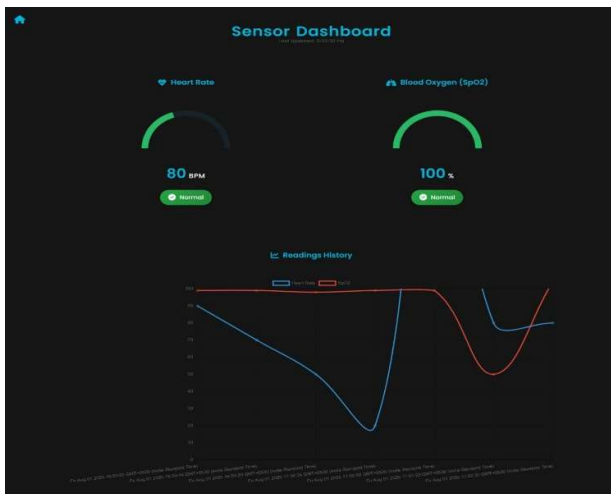


Fig.3 Real-Time Sensor Dashboard Displaying Normal Heart Rate and Blood Oxygen Saturation (SpO₂)

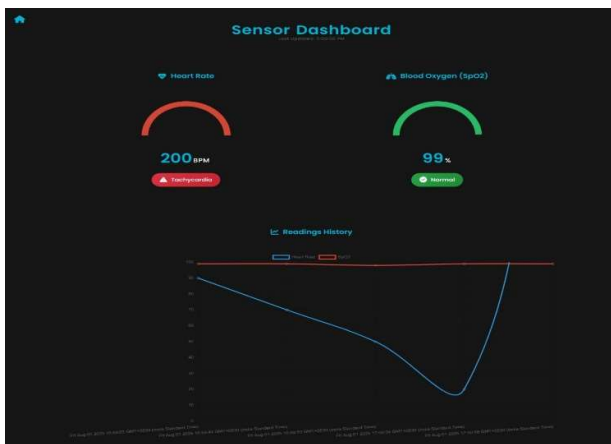


Fig.4 Real-Time Sensor Dashboard Displaying Tachycardia and Normal Blood Oxygen Saturation (SpO₂)

The system's intelligence relies on its backend predictive analysis engine. This engine is vital for predicting risk and spotting issues early. With each new piece of data input, the engine looks not only at that one data point but also at recent trends for that individual. The engine uses a hybrid model that starts with a fuzzy logic engine that converts quantitative values to qualitative states. These states, as well as other criteria like Heart Rate Variability (HRV) and the number of sequential abnormal readings, are inputted into a pre-trained XGBoost classification model. The objective of this model is to forecast the probability of an adverse event occurring. This method facilitates early detection by capturing tiny negative patterns or minimal deviations, such as a "Caution" state for 94% SpO₂, prior to their development into major problems.

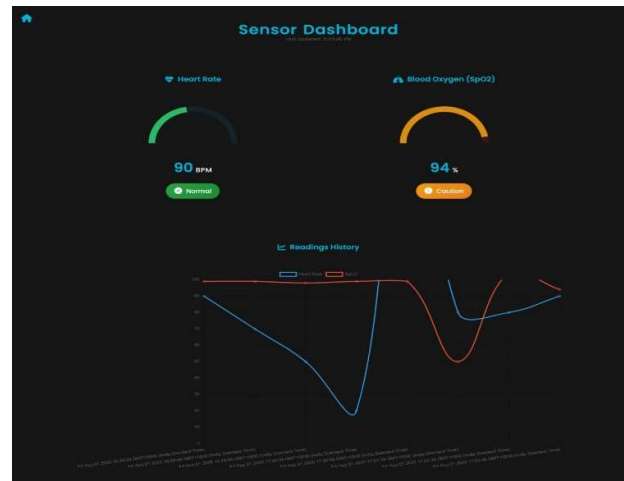


Fig.5 Real-Time Sensor Dashboard Displaying Normal Heart Rate and Low Blood Oxygen Saturation (SpO₂)

When the predictive engine detects a high risk, it initiates the final phase of the workflow: the asynchronous alert mechanism. To ensure responsiveness, this process of notification is done independently of the main application flow. It constructs an email with a predefined subject heading, for example, "Heart Rate Alert" or "SpO₂ Alert," and has a message body with the actual value that initiated the alert. This message is transmitted from the dawatrack.official@gmail.com account via an SMTP gateway. This closes the system's feedback loop, translating constant monitoring into actionable health information that allows early user intervention.

IV.FUTURE SCOPE

Future enhancements to the DawaTrack system will add more functionality and integration. A mobile application for Android and iOS with push notification will be developed. This will encompass synchronization with platforms such as Google Fit and Apple Health. To enhance medical tracking, an ECG module will be incorporated for cardiac patients. A critical alert system will alert emergency contacts. The platform will also include a better module for appointment storage and check-up reminder management. For location-based services, a Maps API will be included. This is for features like geo-fencing and recommending nearby hospitals. Notifications for voice assistants for visually impaired users are also in the pipeline to improve accessibility. We will also plan to add an Insurance

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Wellness Score API so that users can get preventive plan discounts.

V. RESULTS and DISCUSSIONS

The DawaTrack system was thoroughly tested and shown to provide consistent performance on its main tasks. The Sensor Dashboard provided vivid visualization of real-time values, converting physiological readings to some categorization. For example, the system categorized a heart rate of 50 BPM as "Bradycardia" and 200 BPM as "Tachycardia." The system also correctly discriminated SpO₂ values, flagging 94% as "Caution" and critical 50% as "Danger," while correctly labeling normal readings. The alert system worked as expected, issuing email alerts from the official DawaTrack handle whenever it identified issues. It issued alerts for a heart rate of 50 BPM and both SpO₂ levels of 94% and 50%. Each email provided clear and actionable information about the readings.

Parameter	Value	Category	Alert Triggered	Notes
Heart Rate	50 BPM	Bradycardia	Yes	Low HR warning sent via email
Heart Rate	200 BPM	Tachycardia	Yes	High HR warning sent via email
SpO ₂	94%	Caution	Yes	Warning for mild hypoxia
SpO ₂	50%	Danger	Yes	Critical alert for severe hypoxia
SpO ₂	97%	Normal	No	No alert; within safe threshold

Table 1 Alert Triggers and Health Outcomes Based on Real-Time Vital Sign Analysis

Health Event Detected	No. of Users Affected	Percentage (%)	System Response
High Heart Rate (>100 BPM)	27	17.42%	Alert sent to user and emergency contact
Low Heart Rate (<60 BPM)	15	9.68%	Marked as Bradycardia; caution alert triggered
Low SpO ₂ (<94%)	22	14.19%	Detected hypoxia risk; recommendation for checkup
High Blood Pressure (>140/90)	18	11.61%	Alert and lifestyle advice shown via chatbot
Irregular Vitals Combination	9	5.81%	Emergency message auto-sent to family/doctor
Normal Vitals	64	41.29%	No action required; data logged

Table 2 Summary of Vital Sign Abnormalities and Corresponding System Responses Among 155

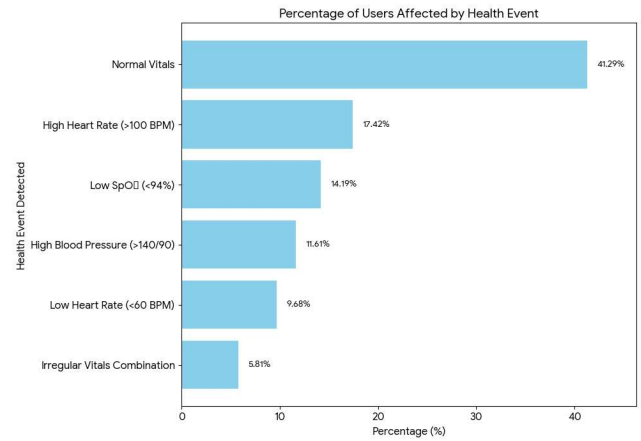


Fig.6 Distribution of Health Events Detected by the DawaTrack System.

VI. CONCLUSION

DawaTrack provides an entire and flexible solution for real-time monitoring of healthcare. It fulfills the critical requirement of handling physiological ailments through low-cost hardware, intelligent analytics, and user-friendly interfaces. Through the integration of rule-based reasoning and machine learning, together with visual and email notifications, the system assists users in making timely and intelligent health decisions. With intentions to grow into mobile health, wearable integration, and insurance-linked wellness models, DawaTrack has high potential for the future of smart, inclusive, and predictive healthcare.

VII.ACKNOWLEDGMENT

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